

CSC343H5 Introduction to Databases
Sprint 4: Normalization
 Student Submission Template

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How to use this template. Write each answer beneath its prompt, replacing the grey *[placeholder]* text. Use $\rightarrow$ for the arrow ($A \rightarrow B$), $\text{clo}\{\dots\}$ for a closure ($\{A\}^+$), and $\text{pk}\{\dots\}$ to underline a primary-key attribute. Show every step; closures and decomposition steps are where the marks are.

Reference: research_flat. Key (grant_code, project_id, researcher_id). Attributes: grant_code, grant_title, funding_body, funding_sector, project_id, project_title, budget, researcher_id, researcher_name, institution, role_on_project. Given FDs:

- FD1: grant_code $\rightarrow$ grant_title, funding_body
- FD2: project_id $\rightarrow$ project_title, budget
- FD3: researcher_id $\rightarrow$ researcher_name, institution
- FD4: funding_body $\rightarrow$ funding_sector
- FD5: grant_code, project_id, researcher_id $\rightarrow$ role_on_project

1. Task 1: Closure & Candidate Key

(a) Closure of the composite key

Let $K = \{\text{grant_code}, \text{project_id}, \text{researcher_id}\}$. Show each step; the final set must contain every attribute.

$$\begin{aligned}
 K^+ &= K && \text{(start)} \\
 &= [\text{add attributes}] && \text{(by [FD?])} \\
 &\vdots \\
 &= [\text{all attributes}] && \text{(done)}
 \end{aligned}$$

(b) Candidate key

[State whether the composite key is the only candidate key (or list any others), and justify in one sentence, e.g. which attributes never appear on a right-hand side and so must be in every key.]

(c) Prime and non-prime attributes

Prime: [...] **Non-prime:** [...]

2. Task 2: Minimal Cover

Step 1: Single-attribute right-hand sides

[Rewrite every FD so each has one attribute on the right.]

Step 2: Remove redundant left-hand-side attributes

Only FDs with more than one attribute on the left need checking. For each, show the closure test.

$$\{[subset\ of\ the\ LHS]\}^+ = \{[...]\} \Rightarrow [attribute\ is\ /\ is\ not\ redundant]$$

[Repeat as needed.]

Step 3: Remove redundant FDs

For each FD $X \rightarrow Y$, compute $\{X\}^+$ *without* that FD; if Y is still reached, the FD is redundant.

$$\{[X]\}^+ \text{ (without } [FD?]) = \{[...]\} \Rightarrow [keep\ /\ drop]$$

[Repeat as needed.]

Resulting minimal cover

$$\text{Minimal cover} = \{ [list\ your\ final\ FDs] \}$$

3. Task 3: BCNF Decomposition

For each step, name the violating FD, give the two resulting relations with their keys, and show the lossless check. Add as many steps as you need.

Decomposition steps

Step 1. Decompose $[relation]$ on $[X] \rightarrow [Y]$.

- **Violation:** $[X]$ is not a superkey of $[relation]$ because $[...]$.
- **Resulting relations:** $[R1(attrs)]$ with key $[...]$; $[R2(attrs)]$ with key $[...]$.
- **Lossless check:** shared attributes = $[...]$; these form a key of $[which\ side]$, so the split is lossless.

Step 2. $[...]$

Step 3. $[...]$

Final BCNF schema

List every relation in schema notation with primary keys underlined, for example:

- $\text{grant}(\underline{\text{grant_code}}, \text{grant_title}, \text{funding_body})$
- $[...]$

BCNF verification

[For each final relation, state that every non-trivial FD in it has a superkey left-hand side.]

4. Task 4: 3NF Check & Justification**(a) Dependency preservation**

[For each FD in your minimal cover, say which single relation can check it. State whether the BCNF schema is dependency-preserving. If not, give the 3NF synthesis result and say which schema you would use and why.]

(b) Anomaly example

[Pick one anomaly type (update, insertion, or deletion) and give a concrete example using the sample rows: which rows are involved and what goes wrong.]

5. AI Usage Declaration

[Declare any AI tool use per the course policy. If none, write: "I did not use any AI tools in completing this sprint."]

- **Tool:** Claude Sonnet 5
 What I asked: [...]
 How I verified / applied it: [...]

6. Self-Check (optional)

- ☐ Closure reaches all 11 attributes.
- ☐ Every minimal-cover step shows a closure test.
- ☐ Each BCNF step has a violation, two relations with keys, and a lossless check.
- ☐ Every final relation is verified BCNF.
- ☐ Name, student number, and UTORid filled in above.